

MOORHEAD, MN, USA

WMO: 722129

Lat: 46.839N

Lon: 96.663W

Elev: 917

StdP: 14.22

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04939

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-13.4	-24.1	1.5	-16.5	-18.4	2.1	-11.7	33.3	7.6	29.7	12.1	5.5	170	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	89.7	73.6	86.0	71.5	82.4	68.8	77.1	85.5	74.7	82.0	72.6	79.9	12.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.6	134.1	82.2	72.3	123.9	79.7	70.3	115.3	77.6	41.1	84.7	38.7	82.5	36.8	80.1	89.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 27.8	24.8	21.8	-23.8	94.6	4.2	2.4	-26.8	96.3	-29.3	97.7	-31.7	99.0	-34.8	100.7		
(5)			-24.2	80.9	4.0	2.9	-27.1	83.0	-29.4	84.7	-31.7	86.4	-34.6	88.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	10.7	12.0	28.3	43.6	57.1	67.2	71.1	68.0	61.2	46.9	31.8	15.7		
	DBStd	23.99	14.26	12.90	13.23	10.55	9.48	6.19	5.69	5.78	8.18	9.15	11.30	12.74		
	HDD50	5035	1219	1065	681	239	40	0	0	0	9	170	549	1063		
	HDD65	8584	1684	1485	1139	644	278	47	14	34	168	568	995	1528		
	CDD50	2463	0	0	8	46	260	515	655	558	345	72	4	0		
	CDD65	537	0	0	0	1	33	112	204	127	54	6	0	0		
	CDH74	4897	0	0	0	14	400	956	1893	1086	490	58	0	0		
	CDH80	1415	0	0	0	1	134	245	598	289	135	13	0	0		
(14) Wind	WSAvg	9.5	9.9	9.7	10.3	11.0	10.9	9.1	7.2	7.1	8.6	10.3	10.0	9.7		
(15) Precipitation	PrecAvg	24.1	0.6	0.6	1.1	1.7	3.2	4.4	3.2	2.6	2.6	2.2	0.9	0.9		
	PrecMax	32.8	1.5	1.5	3.6	3.7	7.6	9.6	8.1	7.2	6.3	5.3	3.9	1.9		
	PrecMin	16.2	0.0	0.1	0.0	0.3	1.3	1.1	0.9	0.9	0.3	0.0	0.0	0.2		
	PrecStd	4.4	0.4	0.4	0.7	0.9	1.7	2.1	1.8	1.3	1.5	1.4	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.8	45.7	66.4	75.1	90.4	90.6	92.8	90.7	89.9	81.3	63.1	45.6		
		MCWB	37.0	41.6	58.0	55.3	64.1	71.9	77.6	75.0	71.2	63.1	52.7	42.6		
	2%	DB	36.4	37.2	59.0	70.1	83.5	85.8	89.8	86.4	83.8	72.6	55.5	38.6		
		MCWB	34.4	35.1	51.7	53.7	62.5	68.1	76.2	73.5	68.3	58.3	47.4	35.3		
	5%	DB	34.1	35.6	52.1	64.2	78.8	82.0	86.1	82.3	80.7	65.8	52.0	35.8		
		MCWB	32.9	34.0	46.7	51.1	60.8	66.3	73.9	70.3	66.4	55.7	46.2	33.8		
	10%	DB	30.3	30.3	45.4	61.1	72.8	80.6	82.4	80.8	75.1	61.2	46.0	32.4		
		MCWB	29.8	29.7	42.1	49.1	58.2	65.4	70.9	68.8	64.4	51.9	41.2	31.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.6	41.9	61.0	60.0	71.0	76.8	81.6	78.1	74.7	64.7	55.8	42.8		
		MCDB	39.8	45.9	64.7	70.1	79.0	85.6	87.2	85.8	82.5	75.2	62.6	45.9		
	2%	WB	34.6	36.0	52.2	55.8	67.0	73.3	78.3	75.7	71.8	60.4	49.6	36.1		
		MCDB	35.9	37.4	57.7	65.4	76.7	80.9	86.7	83.0	81.1	70.7	54.6	37.9		
	5%	WB	32.7	33.1	46.2	53.1	64.1	70.6	75.8	73.5	68.7	56.8	45.7	33.6		
		MCDB	33.9	34.7	50.7	62.0	72.3	77.9	83.4	80.2	76.0	64.1	50.6	35.3		
	10%	WB	29.9	29.9	41.8	50.4	61.3	68.1	73.5	71.2	66.4	53.3	41.8	31.1		
		MCDB	30.5	30.7	46.4	58.9	69.1	75.6	80.7	77.8	74.0	59.3	46.2	32.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	16.4	16.1	18.7	20.7	18.8	19.9	20.4	20.5	17.0	15.2	14.6		
		MCDBR	17.5	17.1	23.9	26.9	27.5	22.9	22.6	23.0	25.4	25.5	21.4	18.5		
	5% WB	MCWBR	15.9	14.8	16.5	14.6	13.1	11.0	11.7	11.7	12.3	14.0	14.3	15.6		
		MCWBR	16.9	17.1	22.0	23.1	21.2	18.4	19.9	20.1	21.1	22.4	20.4	16.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.301	0.303	0.306	0.360	0.378	0.391	0.392	0.395	0.359	0.317	0.287	0.276		
		taud	2.227	2.276	2.426	2.333	2.347	2.352	2.345	2.335	2.421	2.516	2.503	2.379		
	Ebn at Noon	Ebn at Noon	244	273	293	284	281	277	275	269	270	268	257	246		
		Edh at Noon	26	30	31	37	38	38	38	37	31	24	20	20		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	528	892	1226	1556	1746	1940	2037	1709	1280	799	494	374		
	RadStd	RadStd	56	95	125	128	117	129	83	143	84	114	56	41		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (28 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page